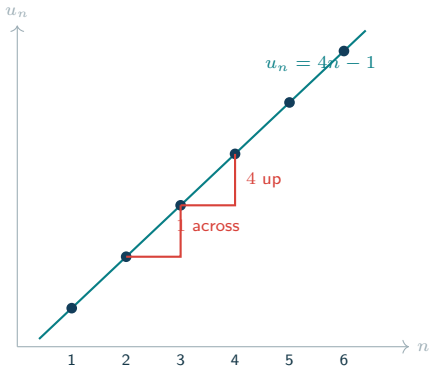
SEQUENCES LADDER 1 OF 9

Sequences, rules and first differences

The step between neighbouring terms is the first difference, and a constant step gives a straight-line rule.

1 / THE RULE AND THE PICTURE

A constant first difference d gives the position-to-term rule $u_n = dn + (u_1 - d)$



A sequence is a list $u_1, u_2, u_3, \ldots$ and its first differences are $\Delta u_n = u_{n+1} - u_n$. When every first difference equals d the terms climb by the same amount each step, so the points lie on a straight line of gradient d, and

$$\boxed{\Delta u_n = d \implies u_n = dn + (u_1 - d)}$$

For $3, 7, 11, 15, \ldots$ the difference is 4 and $u_1 - d = 3 - 4 = -1$, so $u_n = 4n - 1$.

2 / TWO KINDS OF RULE

Add the same step each time, or say the term from its position

sequence	term to term	position to term	first difference
$3, 7, 11, 15, \ldots$	add 4	$4n - 1$	$+4$
$10, 7, 4, 1, \ldots$	subtract 3	$13 - 3n$	-3
$2, 5, 8, 11, \ldots$	add 3	$3n - 1$	$+3$
$\frac{1}{2}, 1, 1\frac{1}{2}, 2, \ldots$	add $\frac{1}{2}$	$\frac{n}{2}$	$+\frac{1}{2}$

3 / FINDING THE RULE IN THREE STEPS

Differences first, then compare with dn

Write down the differences, and if they are all the same number d the rule begins with dn.
Then compare dn with the actual terms and note what must be added or taken away: for $5, 8, 11, 14, \ldots$ the differences are 3, the multiples of 3 are $3, 6, 9, 12$, and each term is 2 more, so $u_n = 3n + 2$.
Check with $n = 1$ and $n = 2$, which give 5 and 8.

4 / THE DIFFERENCE IS A GRADIENT

The first picture of calculus

On the graph of u_n against n the neighbouring points are 1 apart across, so the difference is the rise divided by the run, the gradient of the line between them.
For $4n - 1$ the gradient is 4 everywhere, but for $1, 4, 9, 16$ the differences $3, 5, 7$ keep growing, so the gradient changes and the graph bends, which is what the next sheet measures with second differences.

5 / A TRAP

Two or three terms can fit several rules

The terms $1, 2, 4$ continue as 8 if each is doubled, and as 7 if the differences go $1, 2, 3$, which gives $1, 2, 4, 7, 11$.
A rule found from differences must be tested on more terms than it uses, and a safe habit is to check that the differences really are constant over at least three steps.

6 / TRY THESE

Find the rule

(a) For $7, 10, 13, 16, \ldots$ the difference is 3 and the rule is $3n + 4$. **(b)** For $20, 17, 14, 11, \ldots$ the difference is -3 and the rule is $23 - 3n$. **(c)** Is $2, 5, 10, 17$ a straight-line sequence? The differences $3, 5, 7$ are not constant, so no.

ASIDE / TWO POINTS FIX A LINE

Two terms are enough to find a straight-line rule, and in general $k + 1$ terms fix a polynomial of degree k, as the interpolation sheet shows.

LINKS TO OTHER SHEETS

The next sheet takes the differences of the differences; the polynomial interpolation sheet gives the same rules by Vandermonde and Newton.

Convention. u_n is the nth term; $\Delta u_n = u_{n+1} - u_n$ is the first difference; d is a constant difference.